Yifan Guo

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GitHub: github.com/Ricardo-Guo | Portfolio: Ricardo-Guo.github.io

EDUCATION

Southern University of Science and Technology (SUSTech) Shenzhen, China
B.S. in Robotics *Expected Jun. 2027*
GPA: 3.77/4.00; Average: 90.87/100
Relevant Coursework: Machine Learning, Control Engineering, Robot Modeling and Control, Mechatronics

RESEARCH EXPERIENCE

Agentic Real2Sim2Real for Robot Policy Learning *Aug. 2026 – Nov. 2026*

- Built a closed-loop Real2Sim2Real pipeline for scene reconstruction, policy learning, and physical SO-101 deployment.
- Reconstructed simulation states from dual-view RGB and robot proprioception to align simulation with real execution.
- Trained stage-specific policies from planner-generated simulation data using behavior cloning and domain randomization.
- Deployed fixed-weight policies at 20 Hz with safety guards and automatic stage-level success verification.
- Achieved 16/19 grasps, 13/16 in-tolerance releases, and 12 consecutive verified transitions without human reset.

Vision–LiDAR Fusion for UAV Target State Estimation and Semantic Mapping *May 2026 – Sep. 2026*

- Developed an end-to-end vision–LiDAR system for UAV-target state estimation and semantic point-cloud mapping.
- Implemented STM32-triggered global-shutter camera–Livox Mid360 LiDAR synchronization and extrinsic calibration.
- Integrated Fast-LIVO2 with SegFormer-B0 semantic labeling to generate offline point-cloud maps with per-point labels.
- Used EKF filtering for UAV velocity estimation, achieving 9 Hz output and $<$10 cm error for 95% of results within 5 m.

Metamorphic-Structure Reconfigurable Aircraft *Sep. 2024 – Oct. 2025*

- Co-contributed to a reconfigurable aircraft prototype switching between airship and twin-rotor fixed-wing modes.
- Built dynamics-based Gazebo simulation and debugged Pixhawk–QGroundControl communication for control testing.
- Developed fixed-wing attitude-stabilization and altitude-hold controllers using real-time onboard sensor feedback.
- Implemented Raspberry Pi 4B servo control for attitude adjustment and morphing and completed ground demonstration.

MANUSCRIPTS & INTELLECTUAL PROPERTY

- Y. Huang, **Yifan Guo**, Z. Luo, J. Liu, J. Zhang, H. Lu, and X. Zheng, "Rebuild, Relearn, Re-Execute: Closed-Loop Real-to-Sim-to-Real Policy Learning for Robot Manipulation," submitted to *ICRA 2027*.
- Co-inventor, "Metamorphic Aircraft," Chinese Invention Patent Application No. 2026102057666, filed by SUSTech and accepted by CNIPA; 3rd of 8 listed inventors, 2026.

SELECTED HONORS, AWARDS & GRANTS

- National-level Funded Project, College Student Innovation and Entrepreneurship Training Program, 2026.
- National Third Prize, National College Student Mechanical Innovation Competition, 2026.
- Provincial First Prize, Guangdong Engineering Maker Competition; team ranked 2nd overall in category, 2025.
- SUSTech Climbing Program Research Grant, RMB 20,000, 2025.

TECHNICAL SKILLS

Programming & Tools	Python, C++, PyTorch, Linux/Ubuntu, Git, Docker, ROS 2, MuJoCo, Isaac Gym, Gazebo
Algorithms	Reinforcement learning, imitation learning, vision-language-action model, world model, deep learning, SLAM, 3D reconstruction, PID control
Hardware & Deployment	NVIDIA Jetson, STM32, Raspberry Pi, Arduino, Pixhawk, sensor synchronization
Fabrication	SolidWorks, CAD modeling, 3D printing, mechanical assembly, prototype integration